

A review on the state-of-the-art technologies of electric vehicle, its impacts and prospects



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ABSTRACT

Electrifying transportation is a promising approach to alleviate the climate change issue. The adoption of electric vehicle into market has introduced significant impacts on various fields, especially the power grid. Various policies have been implemented to foster the electric vehicle deployment and the increasing trend of electric vehicle adoption in the recent years has been satisfying. The continual development of electric vehicle power train, battery and charger technologies have further improved the electric vehicle technologies for wider uptake. Despite the environmental and economical benefits, electric vehicles charging introduce negative impacts on the existing network operation. Appropriate charging management strategies can be implemented to cater for this issue. Furthermore, electric vehicle integration in the smart grid can bring many potential opportunities, especially from the perspective of vehicle-to-grid technology and as the solution for the renewable energy intermittency issue. This paper reviews the latest development in electric vehicle technologies, impacts of electric vehicle roll out and opportunities brought by electric vehicle deployment.

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1. Introduction

The emissions of greenhouse gases (GHG) are the unwanted by-product usually associated with burning of fossil fuel for energy needs. The severity of climate change due to GHG emissions has reached a threatening level and can be easily noticeable with the present global warming and melting of large icebergs. Immediate preventive actions and climate policies are needed to slow down the worsening climate change impacts. In response to this matter, International Energy Agency (IEA) has outlined scenarios for future energy system to limit the average global temperature increase to two degree Celsius by 2050 [1]. The amount of GHG emissions is projected to double by 2050 if no initiatives are taken to cater for this situation [2]. Transportation sector accounted for one quarter of energy-related GHG emissions in 2009.

Various efforts are being undertaken to reduce the emissions from the transportation sector. The focus is to develop new fuels and introduce clean technology features for vehicles, with the aim to reduce GHG emissions and also improve vehicle performance. Electrifying transportation is one of the promising approaches with many benefits. Electric Vehicles (EVs) could improve energy security by diversifying energy sources, foster economic growth by creating new advanced industries and most importantly, protect environment by minimizing tailpipe emissions. EVs show a better performance than internal combustion engine vehicles (ICEVs) due to the usages of more efficient power trains and electric motors [3]. Governments around the world are implementing different initiatives, policies and programmes for wider EVs uptake. Incentives to EVs purchase cost, development of charging infrastructure and increase of public awareness of EVs benefits are among the actions taken to promote EVs. The efforts seemed to pay off as EVs start to gain acceptance from public. According to the Global EV Outlook prepared by Electric Vehicle Initiative (EVI) and IEA [4], the global EVs stock is more than 180,000 at the end of 2012. This allows EVs to take up 0.02% of the global vehicle stock and continue to be engaged in research and development.

The continual development of EV technologies is the crucial factor to improve EVs performance and ensure its competitiveness. For instance, development focus has been placed on technologies of power train, battery and charging infrastructure. In order to meet the various demands, different configurations of power train are designed along the EVs development process, such as series, parallel and series-parallel configurations [5]. These power train configurations could improve fuel economy and enhance vehicle driving range due to the use of highly efficient electric motor [3]. Similarly, battery technology is evolved from lead-acid to nickel-based to ZEBRA battery and finally to lithium-based types, in order to find a storage technology which has high energy density, high power density, light in weight, inexpensive, safe and durable [6]. Research has been focused on metal-air battery, which has high energy density up to 1700 Wh/kg that can compete with the performance of the conventional internal combustion engine vehicle [7]. Charging infrastructure which provides high charging power, such as direct current (DC) fast charging station is gradually adopted to replace slow chargers in order to solve the range anxiety issue among EVs drivers [8].

Studies are being carried out to analyze the impact of EVs take up, with focus on economic, environment and technical issues on power grid, which will be comprehensively investigated in Section 4. The cost impact of EVs is highly dependent on the generation mix used for EVs

charging [9]. As EVs rely on electricity from power grid to propel, the cost of power generation strongly affects the cost of EV usage. The economic impact of EV deployment can be evaluated from the viewpoint of power grid and viewpoint of EV owners. Power grid need to have more generation capacity for the additional EV load demand while EV owners have to pay the high initial purchase cost of EV at the present time. However, with the implementation of coordinated charging, energy trading and various electricity rates policy, EV deployment can be profitable for the operation of power grid and EV owners. The environmental impact of EVs roll out is subjective [10,11]. The obvious observation is EVs have zero tailpipe emissions, which is clean and green to environment. However, EVs use electricity generated from power grid and the process of electricity generation does produce GHG emissions. Hence, the environmental impact of EVs usage depends on the sources of electricity. Renewable energy is widely employed lately and has favoured EVs to be more environmental friendly than conventional ICEVs [12]. The interconnection of EVs to power grid to receive charges raise concerns about the negative impacts of EVs charging on power grid. Based on extensive search on literature, the anticipated problems associated with EVs charging are harmonics, system losses, voltage drop, phase unbalance, increase of power demand, equipment overloading and stability issues [13,14]. The impact of EV charging on power grid will be further discussed in Section 4.3.

The aim of this paper is to review the current EV status, the development of EV technologies, the impacts of EV deployment and future potentials brought by EV field. The paper is organized into several sections. Section 1 presents the introduction. Section 2 describes a brief history of EVs and its current status. Section 3 gives an insight on the current EV technologies, particularly on power train, battery and charging of the battery. Impacts of EV deployment are explained in Section 4. The future technologies and potentials of EV deployment in smart grid are shown in Section 5. Research gaps and limitations in present EV field are demonstrated in Section 6 and Section 7 concludes the paper.

Table 1
EV status.

Country	EV stock [17]	EVSE stock [17]	National EV target [20–23]
United States	71,174	15,192	1,000,000 by 2015
Japan	44,727	5009	20% of total vehicle sales by 2020
France	20,000	2100	2,000,000 by 2020
China	11,573	8107	500,000 by 2015
United Kingdom	8183	2866	1,500,000 by 2020
Netherlands	6750	3674	1,000,000 by 2025
Germany	5555	2821	1,000,000 by 2020
Portugal	1862	1350	750,000 by 2020
India	1428	999	6,000,000 – 7,000,000 by 2020
Denmark	1388	3978	200,000 by 2020
Sweden	1285	1215	6000 by 2015
Spain	787	705	250,000 by 2014
Finland	271	2	80,000 by 2020

EV Stock: Cumulative EV stocks by 2012.

EVSE Stock: Non-Residential slow and fast EVSE stocks by 2012.

2. EVs history and current status

EVs have experienced tremendous changes from the nineteenth century until present days. It is surprising to learn that EVs were once the top choice for transportation use. The beginning of electric-powered vehicles was favoured by the invention of electric motor. In between years 1832 and 1839, the first prototype electric-powered carriage, which powered by non-rechargeable primary cells was invented by Robert Anderson [15]. After that, different electric-powered carriage prototypes were invented but unfortunately, all of them were not suitable for practical development due to the lack of efficient electric motor and practical rechargeable battery.

In between 1856 and 1881, direct current (DC) electric motor and rechargeable battery went through a series of developments. DC electric motor with high efficiency was developed and most of the credits were given to Werner Siemens, Antonio Pacinotti and Zénobe Gramme [15]. In addition, the first practical rechargeable lead-acid battery was invented by Gaston Planté in 1859 and was improved into marketable product by Camille Alphonse Faure around year 1881 [16]. The technology advancement of DC electric motor and rechargeable battery provided a major boost to EV industry. For instance in New York City, the first commercial EV introduced into market was an electric taxi in 1897. In just three years, EVs took up 28% of the road vehicles and was the favored vehicle of choice [17].

However, EVs faced a great challenge after a decade. In 1908, gasoline-powered vehicles such as Ford Model T were brought into market by Henry Ford. In 1912, the invention of electric starter by Charles Kettering removed the need of hand crank to start the gasoline-powered vehicles. In addition, the availability of cheap petrol resulted in lower usage cost of gasoline-powered vehicles compared to EVs. On the other hand, EVs could travel relatively shorter distances and only limited charging stations were available [18]. Due to all these factors, gasoline-powered vehicles received great acceptance and EVs faded out of popularity. Around year 1935, there was not a single EV on the road.

A few decades later, the emissions issue of gasoline-powered vehicles and high oil price had renewed interests in EVs. Governments had implemented regulatory actions to reduce air emissions and promote electric and hybrid vehicles development. One among these regulatory actions was California's Zero Emission Vehicle Mandate in 1990, which required two percent and ten percent of the total vehicles to have no emission by 1998 and 2003, respectively [19]. Many

automakers put efforts in producing hybrid vehicles. In 1996, General Motors produced and leased the EV1 model [17]. The next year, Toyota introduced the world's first commercial hybrid electric vehicle (HEV), Prius in Japan and 18,000 units were sold in the first production year [19]. As the oil price kept increasing, more automakers were committed to vehicle electrification. From year 2010 onwards, battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs), such as Nissan Leaf, Mitsubishi i-MiEV, Chevrolet Volt and Tesla Model S have started to enter into automotive industry.

Based on the Global EV Outlook prepared by EVI and IEA, the global EVs stock is more than 180,000 at the end of 2012 [4]. Table 1 shows the current status of EV around the world. The countries' EVs stock and number of Electric Vehicle Supply Equipment (EVSE) installed in 2012, as well as the national EVs target are depicted in Table 1. In this context, EVs are defined as passenger car BEVs, PHEVs and fuel cell electric vehicles. EVSE is defined as non-residential slow and fast chargers. United States has the largest number of EVs on the road, which takes up more than 70,000 units or 38% of the global EVs stock. The number of EVSE installed are more than 15,000 units. Japan is the second largest EV country, which takes up 24% of the global EVs stock. European countries take up approximately 11% of the total global EVs stock. Countries around the world place high national EV target to be achieved in near future. Various policies and actions have been taken to meet the target, such as incentives to EVs purchase cost and charging infrastructure development.

In the Malaysia, EVs and the development of related infrastructure are promoted by the National Automotive Policy [24]. Ministry of Energy, Green Technology and Water Malaysia has appointed GreenTech Malaysia to develop a roadmap for EV deployment in Malaysia. One of the actions taken was the implementation of an EV pilot project in 2011, which was the Fleet Program Test Vehicle (FTV) in Putrajaya and Cyberjaya [25]. The FTV implementation will ensure that the provision of EV roadmap is comprehensive and promote EVs to the public. In 2013, Mitsubishi i-MiEV and Nissan Leaf were brought into the Malaysian market as the very first two EVs in Malaysia. Public charging stations were also installed for the convenience of EV drivers. For instance, First Energy Network built up EV charging stations at Bangsar shopping centre, Suria Kuala Lumpur Convention Centre, Lot 10 shopping centre and Petronas Solaris in Serdang [26]. Recently, a project named Electric Bus 1 Malaysia (EB1M) by Sync R&D was implemented to promote electrically-propulsion concept into public transportation. The goal is to put 2000 of these electric buses on Malaysian roads by 2020 [27].

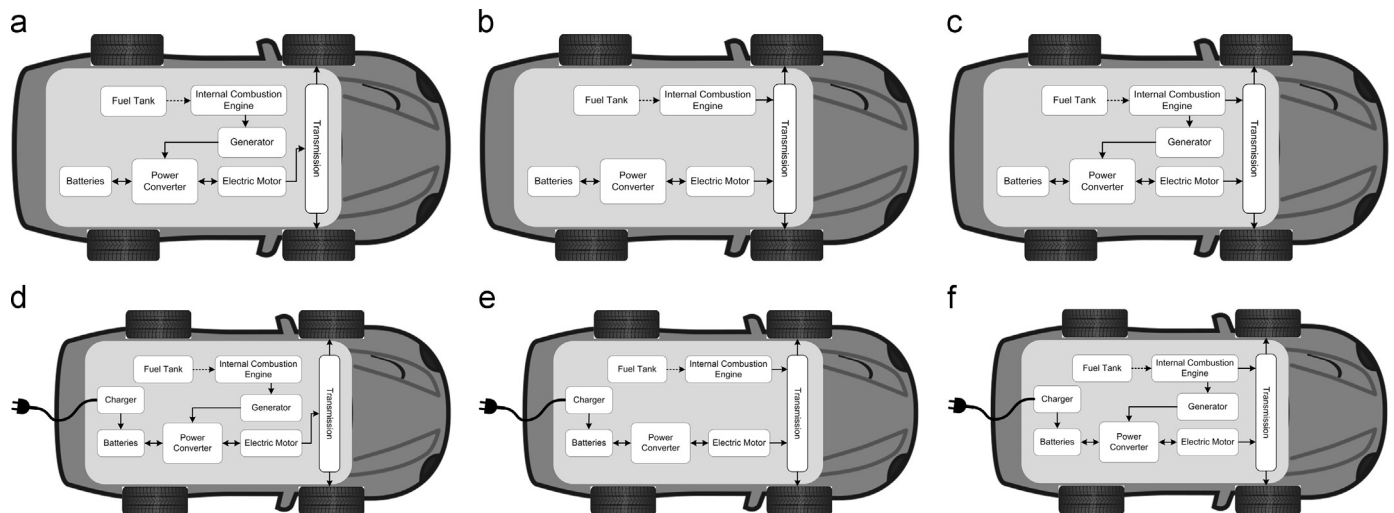


Fig. 1. Power train configurations: (a) Series HEV, (b) Parallel HEV, (c) Series-parallel HEV, (d) Series PHEV, (e) Parallel PHEV, (f) Series-parallel PHEV.

3. EVs technologies

EVs went through a series of technological developments before gaining the recent popularity. The continual development of EV technologies is important in order to compete with the dominant ICEVs and for wider EV deployment. Attentions have been placed on improving technologies, especially the power train, battery and charging infrastructure. Consequently, these components experience major shift along the EV development process. Different configurations of power train designs are available, such as series, parallel and series-parallel configurations. Similarly, battery technology is transformed from lead-acid to nickel-based to ZEBRA battery and finally to lithium-based types. There are also many potential battery types, such as metal-air battery, which is comparable with conventional internal combustion engine vehicle in term of energy density. Charging infrastructure which provides fast charging facility is adopted into market lately to solve the short-coming of long recharging time of the common slow chargers.

3.1. Power train

EVs can be categorized into few types based on the vehicle hybridization ratio, which are hybrid electric vehicles (HEVs), plug-in hybrid electric vehicles (PHEVs) and battery electric vehicles (BEVs). HEVs propel through a combination of an internal combustion engine and an electric motor. HEVs cannot accept charge from external energy sources, such as power grid. Battery is charged by the built-in internal combustion engine or through an energy recovery mechanism called regenerative braking. This mechanism slows a hybrid electric vehicle by converting its kinetic energy into chemical energy, which can be stored in battery for later use. PHEVs are similar to HEVs but with extra features, such as larger battery packs and can be recharged from the distribution grid. Meanwhile, BEVs operate purely on the battery power.

Both HEVs and PHEVs propel by using two energy sources. HEVs and PHEVs usually runs on internal combustion engine and electric motor. A few power train configurations are developed for HEVs and PHEVs in order to achieve different objectives, such as improve fuel economy, increase power and minimize cost. The most common power train configurations of HEVs and PHEVs are series, parallel and series-parallel configuration, as depicted in Fig. 1.

As shown in Fig. 1(a), a series HEV or an extended range electric vehicle uses power from electric motor as the only propulsion source. Other than recharging the battery by regenerative braking, an internal combustion engine and a generator are included to recharge the battery whenever the state of charge (SOC) of the battery is low. The electric motor is mechanically attached to the transmission and wheels. Meanwhile, the internal combustion engine is mechanically decoupled from the transmission and not connected to the wheels. The series HEV is suitable for city driving as it can deal with frequent stop-and-run driving pattern. This type of HEV can improve the overall vehicle efficiency to around 25% and is simpler to design, control and implement [5].

A HEV with typical parallel power train configuration is depicted in Fig. 1(b). In a parallel HEV, both the electric motor and internal combustion engine are mechanically coupled to transmission and simultaneously transmit power to turn the wheels for vehicle propulsion. Due to two propulsion sources in a parallel HEV, the overall vehicle efficiency improvement of this HEV type is more than series HEV, approximately 40% more efficient than conventional car [5]. A parallel HEV is suitable for city driving pattern and also highway driving pattern because electric motor and internal combustion engine can complement each other during different driving conditions. Some available parallel HEVs in market now are Honda Insight and Ford Escape.

The series-parallel HEV, as shown in Fig. 1(c), combines the features of series HEV and parallel HEV. Both the electric motor and internal combustion engine are mechanically coupled to transmission and wheels. The series-parallel HEV can run in either series or parallel mode. Despite having the benefits of both series HEV and parallel HEV, the design of a series-parallel HEV is much more complicated and costly. Toyota Prius is an example of a commercial series-parallel HEV.

Similar to HEV, PHEV can have series, parallel and series-parallel power train configuration with additional on-board battery charger, as depicted in Fig. 1(d)–(f). However, PHEV can be plugged-in to power grid and externally charged. PHEV has larger battery pack and can propel in all-electric drive mode for longer period compared to HEV. There are two operating modes for PHEV, which are charge-depleting mode and charge-sustaining mode [28]. PHEV usually operates in charge-depleting mode at vehicle start up, where the vehicle propulsion power comes from battery power. As a result, the state of charge in the battery reduces in charge-depleting mode. When the battery state of charge reaches a predefined threshold, charge-depleting mode is switched to charge-sustaining mode, where the internal combustion engine turns on. Hence, vehicle is powered by two sources and the battery state of charge is maintained at the predefined threshold in charge-sustaining mode.

Fig. 2 shows a typical power train configuration of BEV. BEV uses electric motor as the sole propulsion source. Therefore, BEV has all-electric propulsion system and always operates in charge-depleting mode. The large battery packs can be recharged through regenerative braking on drive and externally charged when the vehicle stops. Since BEV have all-electric propulsion system, the distance it can travel is based on the battery capacity. For instance, Nissan Leaf has a 24 kWh lithium-ion battery packs installed and can travel up to 160 km in a single full charge. Generally, BEV is suitable for city drive as it has limited all-electric drive range. However, BEV has decent benefits, such as zero tailpipe emissions and better vehicle performance.

3.2. Battery

Battery is the core component of an EV and one of the two propulsion sources of HEV and PHEV. Meanwhile, it is the sole propulsion source for BEV. There are still some constraints on present EV battery technology, which becomes the barrier for wider EV uptake. The current EV battery has relatively low energy density, which directly affects the maximum all-electric drive range of the EV.

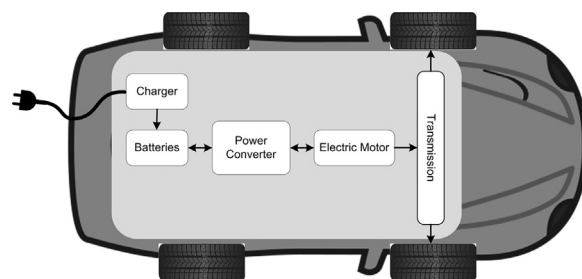


Fig. 2. Typical power train configuration of BEV.

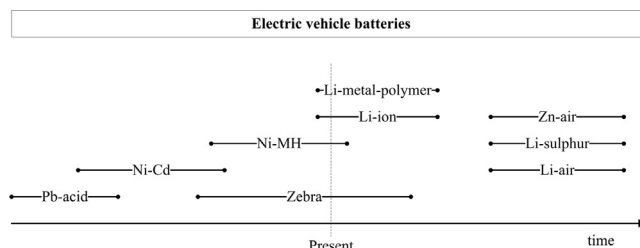


Fig. 3. Development timeline of EV battery.

In addition, high battery cost has put EV at a disadvantage position since the purchase cost of EV is considerably higher than an conventional internal combustion engine vehicle. There are also concerns about the battery life cycle and its safety features.

However, EV battery did go through tremendous improvements in the past decades. EV battery technology went through a few development phases in order to invent the battery with high energy density, high power density, inexpensive, safe and durable. Fig. 3 shows the timeline of EV battery development. The initial battery technology used in transportation was lead-acid battery. The name of lead-acid comes from the combination of lead electrodes and acid used to generate electricity. Lead-acid battery is a matured technology and cheap. However, there are few apparent drawbacks of lead-acid battery, such as low energy density, heavy, require inspection of electrolyte level and is not environmentally friendly.

Lead-acid battery was soon replaced by nickel-based battery, such as nickel-cadmium (Ni-Cd) and nickel-metal hydride

(Ni-MH). Nickel-based battery is considered as relatively matured technology and have higher energy density when compared to lead-acid battery. Most of the commercial EVs in the past decade used nickel-based battery, especially Ni-MH battery for propulsion. However, this type of battery technology has significant drawbacks, such as poor charge and discharge efficiency, high self-discharge rate, has memory effect and poor performance in cold weather. The memory effect of Ni-Cd battery proved it is not suitable for EV application, which requires high charge and discharge rate. In fact, Ni-Cd battery has been banned due to toxicity of its components [6]. The characteristics of poor charge efficiency and very high self-discharge rate up to 20% per month has limited the use of Ni-MH battery in future EV application since it takes longer recharging time and discharges itself even when the battery is not in use.

ZEBRA battery or sodium-nickel chloride (Na-NiCl₂) was introduced into EV field about the same time as Ni-MH battery. This type of battery uses sodium salt as electrolyte and has an

Table 2
Comparison of EV battery types [5,7,30,32–34].

Battery type	Nominal voltage (V)	Energy density (Wh/kg)	Volumetric energy density (Wh/L)	Specific power (W/kg)	Life cycle	Self discharge (% per month)	Memory effect	Operating temperature (°C)	Production cost (\$/kWh)
Lead acid (Pb-acid)	2.0	35	100	180	1000	< 5	No	-15 to +50	60
Nickel-cadmium (Ni-Cd)	1.2	50–80	300	200	2000	10	Yes	-20 to +50	250–300
Nickel-metal hydride (Ni-MH)	1.2	70–95	180–220	200–300	< 3000	20	Rarely	-20 to +60	200–250
ZEBRA	2.6	90–120	160	155	> 1200	< 5	No	+245 to +350	230–345
Lithium-ion (Li-ion)	3.6	118–250	200–400	200–430	2000	< 5	No	-20 to +60	150
Lithium-ion polymer (LiPo)	3.7	130–225	200–250	260–450	> 1200	< 5	No	-20 to +60	150
Lithium-iron phosphate (LiFePO ₄)	3.2	120	220	2000–4500	> 2000	< 5	No	-45 to +70	350
Zinc-air (Zn-air)	1.65	460	1400	80–140	200	< 5	No	-10 to +55	90–120
Lithium-sulfur (Li-S)	2.5	350–650	350	–	300	8–15	No	-60 to +60	100–150
Lithium-air (Li-air)	2.9	1300–2000	1520–2000	–	100	< 5	No	-10 to +70	–

Table 3
SAE charging levels [35,36].

Charging level	Charging rating	Charging time	Remark
AC level 1	120 V, 1.4 kW (12 A) 120 V, 1.9 kW (16 A)	PHEV: 7 h (SOC–0% to full) BEV: 17 h (SOC–20% to full)	On-board charger
AC level 2	240 V, up to 19.2 kW (80 A)	For 3.3 kW charger: PHEV: 3 h (SOC–0% to full) BEV: 7 h (SOC–20% to full) For 7 kW charger: PHEV: 1.5 h (SOC–0% to full) BEV: 3.5 h (SOC–20% to full) For 20 kW charger: PHEV: 22 min (SOC–0% to full) BEV: 1.2 h (SOC–20% to full)	On-board charger
AC level 3 ^a	> 20 kW, single phase and three phase	To be determined	To be determined
DC level 1	200–450 V _{DC} , up to 36 kW (80 A)	For 20 kW charger: PHEV: 22 min (SOC–0% to 80%) BEV: 1.2 h (SOC–20% to full)	Off-board charger
DC level 2	200–450 V _{DC} , up to 90 kW (200 A)	For 45 kW charger: PHEV: 10 min (SOC–0 to 80%) BEV: 20 min (SOC–20 to 80%)	Off-board charger
DC level 3 ^a	200–600 V _{DC} , up to 240 kW (400 A)	For 45 kW charger: BEV (only): < 10 min (SOC–0 to 80%)	Off-board charger

Voltages are nominal configuration voltages, not coupler ratings. Rated power is at nominal configuration operating voltage and coupler rated current. Ideal charge times assume 90% efficient chargers, 150 W to 12 V loads and no balancing of traction battery pack.

BEV charging always starts at 20% SOC, faster than a 1 C rate and stop at 80% SOC instead of 100%. PHEV can start from 0% SOC since the hybrid mode is available.

^a Not finalized.

extremely high operating temperature ranges from 245 to 350 °C. ZEBRA battery has high energy density and power density, which is suitable for EV application. However, the extreme operating temperature has placed much pressure on its thermal management and safety concerns [29].

The introduction of lithium-based battery as EV battery shifts EV to a new era. Lithium-based battery is one of the promising battery technologies with high energy density, high power density, light, cheap, non-toxic and can accept fast charge. Lithium-based battery dominates the most recent group of EV. For instance, lithium-ion battery packs are used in Nissan Leaf, Mitsubishi i-MiEV, Tesla Model S and Chevrolet Volt, which are the top EV choices at the present time. A few established batteries fall under the category of lithium-based battery, such as lithium-ion (Li-ion), lithium-ion polymer (LiPo) and lithium-iron phosphate (LiFePO₄). LiPo battery evolved from Li-ion battery, which can be shaped into various sizes for better packaging optimization. Meanwhile, LiFePO₄ battery offers high power density, more life cycle and better safety but with the drawbacks of lower energy density if compared to Li-ion battery [30]. The current lithium-based battery has some limitations, especially on where battery malfunction can lead to fire risk and explosion [31]. The present lithium-based battery technology is not fully matured but definitely hold the potential to be the perfect rechargeable battery for future EV application.

There are some battery technologies in the experimental phase, which gives superior performance. These batteries are lithium-sulfur (Li-S), zinc-air (Zn-air) and lithium-air (Li-air). Li-S battery has relatively high energy density in the category of lithium-based battery and has apparent advantage of low cost due to the use of inexpensive sulfur. However, Li-S battery has high discharge rate and short life cycle [32]. The other potential candidate for future EV battery is Zn-air

battery. This kind of battery has very high energy density, which is higher than lithium-based battery. The main drawbacks of the current Zn-air battery are low power density and short life cycle. Similarly, Li-air battery is still in the prototype stage and has not been commercialized yet. However, the theoretical high energy density of more than 1700 Wh/kg allows it to compete with the conventional internal combustion engine vehicle [7]. Present research has been focused on the development of this attractive battery technology in order to extend the all-electric drive range of EV.

Table 2 shows the comparison of available battery technologies for EV application. The characteristics of the batteries, such as nominal voltage, energy density, specific power, life cycle, percentage of self discharge per month, memory effect, operating temperature and production cost per kWh have been numerically revealed in details.

3.3. Charger

In PHEV and BEV, the battery packs can be recharged externally from the power grid through the charger device. A charger is required in the EV battery charging process because the power grid supply is in the alternating current (AC) form while the battery is in direct current (DC) form. The EV charger is designed to rectify the AC power level from grid to suitable DC power level for EV battery charging. In order to perform this task, an EV charger is usually constructed as an AC/DC converter or rectifier. In some cases, for instance the fast charging station, an additional DC/DC converter is included in the design of the EV charger for better energy conversion. EV chargers can be installed on-board and off-board of the vehicles. On-board charger is often designed in small size to reduce the weight burden for EV. It also has low power rating and mainly to be used for slow charging. On the

Table 4
Converter topologies of EV chargers.

Charger concepts	Converter topologies	Objectives/features	References
Front-end AC/DC converter and back-end DC/DC converter	Front-end AC/DC converter: active rectifier with filter	(a) decoupled PQ control (b) DC-link voltage regulation (c) unity power factor correction (d) input current harmonic elimination (e) reactive power compensation	[42–45] [42,46,47] [43,48–51] [47–49,51,52] [45,47,53]
	Back-end DC/DC converter: (a) full bridge converter with phase shifted zero voltage zero current switching	(a) achieve zero voltage and zero current switching for both leading-leg and lagging-leg switches; reset primary current during freewheeling period; provide galvanic isolation	[42,43,46,50,51]
	(b) half-bridge	(b) can operate either in buck or boost mode; low switching and conduction losses due to low current ratings of active component; lesser number of passive component; high efficiency	[5,44,47]
	(c) CUK	(c) low input and output current ripples	[5,47]
	(d) SEPIC/Luo	(d) improve the Cuk converter by having small voltage rating in a transfer capacitor	[5,47]
	(e) cascaded	(e) low electrical and thermal stress	[5,47]
	(f) interleaved	(f) reduce current ripples	[5,47,49]
	(g) buck/boost converter	(g) simple charging operation	[48,52]
	Front-end AC/AC device: (a) three phase transformer	(a) change input supply voltage to appropriate voltage level; provide galvanic isolation	[54–57]
	(b) Z-source network	(b) acts as static transformer; can reduce harmonic current and inrush current; simple structure and control	[58]
Use of existing electric drive components as charger (integrated charger)	Back-end AC/DC converter: (a) three phase three level diode clamped converter	(a) allow bi-directional power flow; provide ancillary services to the grid	[54]
	(b) active rectifier	(b) achieve unity power factor; implement fast charging	[55–57]
	(c) uncontrolled rectifier	(c) provide simple rectification process of AC input to DC output	[58]
	(a) inverter of electric drive as battery charger (b) the inductors of electric drive motor as filter	(a) better efficiency, cost saving; weight and space optimization (b) reduce current ripple; reduce harmonic; reduce system components; save cost, weight and space	[59–65] [60–65]
Wireless charger (inductive charger)	Consists of transmitting unit (rectifier, high frequency inverter and feeding coil) and pickup unit (receiving coil and rectifier)	Safe; durable; reduce vehicle weight; moderate efficiency	[66–69]

other hand, off-board EV charger is built at dedicated locations to provide fast charging service.

3.3.1. Charging standards

There are a few established international standards for EV charging, such as Society of Automotive Engineers (SAE), International Electromechanical Commission (IEC) and CHAdeMO EV standards [21]. The standards are published by the countries with most EV stocks at the present time, which are America, European Union and Japan.

3.3.1.1. SAE EV standard. Table 3 shows the SAE charging levels and rating terminology with reference to SAE Electric Vehicle Conductive Charge Coupler Standard (SAE J1772) [35,36]. SAE charging standard is categorized into a few levels, which are level 1, level 2 and level 3 for both AC and DC. In this standard context, AC charging are done by using the on-board charger of the EV. Meanwhile, DC charging is performed through the off-board EV supply equipment, which is the dedicated charging station installed at fixed location.

AC level 1 charging is designed for slow charging from a 120 V_{AC} single-phase power network and suitable for overnight charging. The estimated charging time is up to 17 h to charge a BEV from SOC of 20% to fully charged. The rating terminology of AC level 2 charging is 240 V_{AC} with charging current up to 80 A and charging power up to 19.2 kW. The estimated charging time is as short as 22 min to fully charge a fully depleted PHEV by using the 20 kW charger. On the other hand, DC level 1 charging and DC level 2 charging are used to charge an EV through DC power. An off-board EV supply equipment is used to rectify the supply from power grid into DC output before charging the EV battery. DC level 1 charging has charging power up to 36 kW with 200–450 V_{DC} and charging current up to 80 A. Meanwhile, DC level 2 charging has charging power up to 90 kW with 200–450 V_{DC} and charging current up to 200 A. As shown in Table 3, DC charging can charge up an EV to SOC of 80% within half an hour. Both AC level 3 and DC level 3 charging levels are not finalized yet. The proposed charging power for AC level 3 is more than 20 kW and up to 240 kW for DC level 3.

3.3.1.2. IEC EV standard. While SAE EV charging standard uses “levels” to categorize charging ratings, IEC EV standard uses “types” and “modes” for charging standardization. With reference to IEC 62196-2, this standard classifies EV connecting systems into a few types [37]. In addition, four modes of EV charging from external power grid are introduced via IEC 61851-1 [38]. The first three modes are for AC charging and the estimated charging time to fully charge an EV battery are between three and ten hours. The fourth mode charging is for DC charging and could fully charge an EV within ten minutes.

3.3.1.3. CHAdeMO standard. CHAdeMO standard is introduced as the DC fast charging standard designed for modern EVs to accelerate EV deployment and solve range anxiety issue among EV drivers [39]. CHAdeMO DC fast charging can recharge an EV battery to 80% SOC within 30 min by using the optimal DC charging power of 50 kW (around 500 V and 100 A) [40]. This rating of charging power is claimed to be optimum and balanced in terms of cost and benefits. The DC fast charging is performed via an external dedicated EV charging equipment, which is built in permanent locations. CHAdeMO standard was published as Japanese national standard since October 2012. This standard will be published at IEC EV standard as DC charging standard in the beginning of 2014. The inclusion of CHAdeMO standard in IEC EV standard will be IEC 61851-23 for charging system, IEC 61851-24 for communication and IEC 62196-3 for connector [41].

3.3.2. Converter topologies of EV charger

Extensive literatures on EV charger configurations show that the most common charger configuration consists of the front-end AC/DC converter and the back-end DC/DC converter. Other charger concepts are the front-end AC/AC device with back-end AC/DC converter, integrated charger which uses the existing electric drive components for charging operation and the inductive charging which is consisted of converters and coils. Table 4 shows the summary of various converter topologies of EV chargers. Each charger configuration can achieve specific objectives and will be discussed in detail.

As mentioned earlier, the charger configuration with the front-end AC/DC converter and back-end DC/DC converter is the most ordinary charger configuration. In general, the former converter is used to rectify the input supply in AC form to DC form, while the latter converter is utilized to charge the EV battery. The front-end AC/DC converter is usually an active rectifier in combination with filter device. This controllable converter can achieve different functions, such as decoupled real power and reactive power control [42–45], DC-link voltage regulation [42,46,47], unity power factor correction [43,48–51], current harmonic elimination [47–49,51,52] and reactive power compensation [45,47,53]. The front-end AC/DC converter can attain multiple objectives by manipulating the control system strategy, which will generate the desired pulses for the switching operation of the active rectifier. Some widely used control strategies are direct-power control, current control and direct-voltage control.

Unlike the front-end AC/DC converter, the back-end DC/DC converter has many different converter topologies. One of them is the full bridge converter with phase shifted zero voltage zero current switching (ZVZCS) [42,43,46,50,51]. This type of converter topology can achieve zero voltage and zero current switching for both leading-leg and lagging-leg switches. As it uses transformer in the converter configuration, the full bridge converter with phase shifted ZVZCS also provides galvanic isolation between input and output [43]. This kind of converter topology also can reset the primary current during freewheeling period to reduce the circulating conduction loss [46]. The full bridge converter with phase shifted ZVZCS can achieve high efficiency under various load conditions and output voltage range. The other converter topology of back-end DC/DC converter is the half-bridge design, which is mentioned in [5,44,47]. Half-bridge converter topology can operate either in buck or boost mode based on the switching of the two active switches. It has low conduction and switching losses due to the low current ratings of active components, which leads to high efficiency. The CUK converter has the interesting feature of low current ripples. However, it has high voltage rating on transfer capacitor, which is further improved by the SEPIC/Luo converter topology [5]. In addition, the cascaded back-end DC/DC converter can provide low electrical and thermal stress, while the interleaved converter topology can reduce current ripples [47,49]. Some studies also use simple buck or boost converter as the back-end DC/DC converter to represent charger in order to get simple charging operation [48,52].

The second widely adopted charger configuration is the front-end AC/AC device with back-end AC/DC converter. This kind of charger configuration uses the front-end AC/AC device to shift the AC input supply to appropriate AC level initially before rectifying to DC level by the mean of back-end AC/DC converter to charge the EV battery. From literatures, the front-end AC/AC device can be represented by two topologies, which are three-phase transformer and Z-source network [54–58]. Other than changing the input supply voltage to appropriate voltage level, the three-phase transformer can provide galvanic isolation between the input and EV. Meanwhile, the Z-source network acts as a static transformer, which can reduce the harmonic current and inrush current. It can be easily implemented because of the simple structure and control [58]. On the other hand, the back-end AC/DC converter of the

second charger configuration can be represented by three converter topologies, which are the three-phase three level diode clamped converter, active rectifier and uncontrolled rectifier. The three-phase three level diode clamped converter allows bi-directional power flow, which can be used to implement the vehicle-to-grid operation for EV application. It can also provide ancillary services to the grid, such as frequency and voltage control [54]. The active rectifier converter topology is basically the controllable rectifier. This type of converter topology can be used to achieve unity power factor, as well as to provide fast charging to EV battery [55–57]. On the contrary, the uncontrolled rectifier can provide simple rectification process but without active control feature [58].

The third charger configuration is an interesting concept by using the existing electric drive components to represent charger. This concept is called integrated charger [59–65]. This concept can be practically implemented because EV is externally charged when the vehicle is stopped. Therefore, the electric drive components can be utilized as EV charger when the vehicle halts. In reverse operation, the inverter of electric drive can work as the rectifier. Hence, the inverter in the electric drive is often modified and used as the EV charger [59–65]. Similarly, the electric motor in the EV power train is also utilized to act as inductors and work as filter device [60–65]. The use of the integrated charger can save cost, space and reduce system weight.

The previous three charger configurations can be categorized into the same group named conductive chargers. Conductive chargers require physical connection between the power supply and EV. On the other hand, an emerging new charger concept called inductive charger involves no physical connection between the EVs and power supply. Induction coils are installed inside the EVs and charging station and electromagnetism concept is used to charge the EVs [66–69]. As the inductive charger is a relatively new and immature technology, it has apparent disadvantage of lower efficiency if compared to conductive chargers. However, inductive charger has some decent benefits, such as safe and durable.

3.3.3. Charging methods

Various charging methods can be used to charge an EV battery. The conventional charging methods are constant current (CC), constant voltage (CV), constant power (CP), taper charging and trickle charging [70]. In addition, advanced charging involves combination of the above methods, such as constant current/constant voltage (CC/CV). Pulse-charging and negative pulse-charging are also good charging strategies for fast charging of an EV battery [71].

CC is a charging method to maintain a constant charging current flow to battery by varying the charging voltage, until the battery voltage reaches a predefined value. On the contrary, CV applies constant charging voltage to the battery by varying the charging current, until the charging current drops to almost zero. CP is a charging method just like its name implies, charging a battery with constant power. Taper charging is done through an unregulated constant voltage source and the charging current reduces in an uncontrolled way due to increase in the cell voltage as the charge builds up [72]. Taper charging method posts danger to battery since it will damage the battery if overcharging happens. Trickle charging is used to charge the battery with small current to compensate for the battery's self discharge.

CC/CV charging method, as depicted in Fig. 4, is the favored choice to fast charge lithium-ion battery, which is the battery type used in most of the modern EVs [70,71]. The operation of CC/CV charging method can be categorized into two major processes. Firstly, the battery is charged through a constant current, where most of the battery capacity is charged in this process. When the battery voltage reaches a predefined value, CC mode is switched to CV mode. Then, the battery is charged with reducing current while charging voltage is kept constant. Even though CV mode is used to charge the remaining

battery capacity, the charging duration takes approximately the same amount of time or more time than CC mode due to a necessary decrease in charging current to help top-off the battery [70,71].

Pulse-charging charges a battery by feeding charge current in pulses. The charging rate can be controlled by changing the width of pulses. The interesting feature of this charging method is a short rest period of 20–30 milliseconds between pulses to stabilize the battery chemical actions [71]. The rest period enables the chemical reaction to keep pace with the charging process and therefore, can reduce the gas formation at the electrode surface. A new pulse-charging method, which is duty-varied voltage pulse-charging strategy was proposed in [73]. Instead of using the constant pulse width, this method detects and supplies the suitable charge pulse with varying pulse width to the battery to increase charge speed and charge efficiency. Meanwhile, negative pulse-charging is a complementary method with pulse-charging. This method applies a very short discharge pulse, during the pulse-charging rest period to depolarize the battery and clear off any gas bubbles built up on the electrode during pulse-charging [71]. Negative pulse-charging is claimed to enhance the overall charging process and prolong the battery lifetime. Fig. 5 gives an insight on the general operations of the pulse-charging and negative pulse-charging.

4. Impacts of EV deployment

Extensive research has been carried out to investigate the impacts of EV deployment. Attentions are placed on three major impacts, which are economic impact, environmental impact and impact on power grid due to EV roll out. The core findings from these three categories will be discussed in detail in the sections below.

4.1. Economic impact

The economic impact of EV deployment can be evaluated from two perspectives, which are from the point of view of power grid and viewpoint of EV owners [74]. From the power grid perspective, EVs are additional loads that need to be plugged-into power grid to receive charging. In order to cope with this massive additional

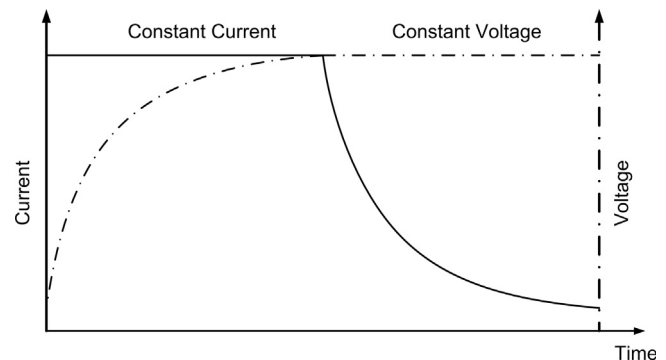


Fig. 4. CC/CV charging profile.

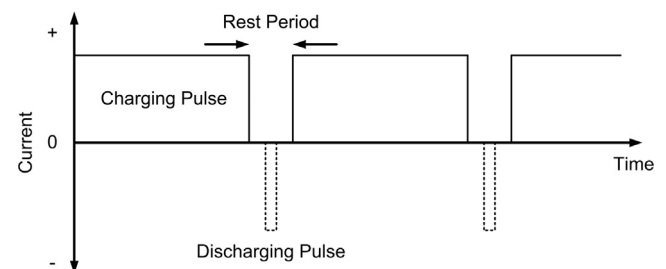


Fig. 5. Pulse-charging and negative pulse-charging.

EV loads, system cost will increase due to increased fuel used for more power generation [75]. There are also more power losses during the power transfer across the power grid to supply these EV loads. However, this situation can be completely changed by manage the EV charging [76]. Controlled EV charging can significantly reduce the system cost with savings up to 60% [77]. The cost reduction is even better with integration of renewable energy resources in power grid, particularly wind energy.

From the EV owners perspective, EVs has low operating cost because of the use of efficient electric motor and inexpensive electricity [78]. However, EV has higher initial purchase cost than conventional ICEVs due to the expensive battery component. A term called “EV payback period” is introduced to estimate the length of time required to recover the investment cost of an EV [79]. Many actions can be implemented to ease the high initial purchase cost of EV, such as mass-producing EVs [80], implement energy trading policy [81] and adopt appropriate charging strategies [82].

In the first glance, the economic impact of EV deployment on both power grid and EV owners are not positive. Power grid need to has more generation capacity for the additional EV load demand while EV owners have to pay the high initial purchase cost of EV at the present time. However, with the implementation of coordinated charging, energy trading and various electricity rates policy, EV deployment can be profitable for the operation of power grid and EV owners.

4.2. Environmental impact

EVs are claimed to be green and environmental-friendly since EVs have zero tailpipe emissions. However, EVs use electricity generated from power grid to charge their batteries and the power generation process does produce GHG emissions. In order to compare the emissions level of EVs to the conventional ICEVs, a parameter called “wells-to-wheels emissions” is introduced. Wells-to-wheels emissions take into account the emissions over the entire life of a vehicle, which includes the energy and materials used to power a vehicle and also the direct tailpipe emissions. Many research conclude that EVs have the lowest wells-to-wheels emissions [12,78,83].

However, EV charging from a power grid with coal-fired and other polluting fuels generation, may cause EVs to have higher wells-to-wheels emissions than ICEVs. For instance, the Texas power grid, which has a mix of coal- and natural gas-fired generation has been shown to yield higher emissions from EVs than ICEVs [84]. Similarly, the Ohio power grid with coal-fired generation yields higher SO₂ and NO_x emissions with EVs use, although a reduced of CO₂ emissions up to 24% from EVs than ICEVs [85]. These results show that EVs could be not environmental-friendly if EVs are charged from a dirty power grid. However, with the wide employment of green renewable energy sources lately, this will make the power grid become greener. Hence, the wells-to-wheels emissions of EVs will be reduced.

4.3. Impact on power grid

EV deployment raises concern about the effect of EV charging on the power grid. The interconnection of large EV fleets to power grid to receive charging can introduce negative impacts to the power grid, such as harmonics, system losses, voltage drop, phase unbalance, increase of power demand, equipment overloading and stability issues [13,14]. Various possible charging rates and dynamic behaviour of EVs even complicate the potential impacts. Therefore, the number of related literature has been increased recently. This section reveals various studies on the grid impact due to EV charging.

4.3.1. Impact on load profile

Numerous studies have been carried out to investigate the impact of EV deployment on grid load profile. Power grids around the world are taken into consideration for this impact study. For instance, impact of EVs on hourly load profile in the United States is carried out in [85]. The results show that if EV owners are allowed to charge their EVs anywhere at any time, EV charging will increase the load at peak hours and late afternoon peak, which are the time of arrival at work and time arrival at home after work, respectively. A delayed charging control is proposed in study [85] to prevent the increase of peak load.

Another study is performed to investigate the impact of EV charging on the load profile of German grid in year 2030 [86]. It reveals that uncontrolled charging of one million of EVs has slight impact on the daily peak load, where the peak load increases only by 1.5%. However, if all the conventional ICEVs in Germany (around 42 million units) are replaced by EVs, then the EV charging will increase the peak load by approximately two times. The study also shows that a maximum peak load reduction of 16% can be achieved with the use of one million EVs as grid stabilizing storages.

In the context of Western Australian power grid, the impact of EV charging on load profile is undertaken in [87]. This study assumes that all new vehicles are EVs to remove the doubt in EV adoption rate. In the uncoordinated charging scenario, the Western Australian power grid can absorb the additional charging loads of 200,000 EVs during the peak demand hour. This study also shows that the grid can accept 900,000 units of EVs without causing any negative impacts on the grid by shifting the EV charging to the night time periods.

Another study was carried out to analyze the effect of large scale EV integration on the Estonian power grid, where the EV penetration level reaches 30% of the total passenger vehicles [88]. The study shows that the EV integration has minor impact on power load profile. For uncontrolled EV charging, the peak load increases by 5%. On the other hand, the peak load will increase only by 4% for the controlled EV charging scenario. In addition, controlled EV charging will disperse the EV loads over the night-time periods and level the load profile.

The impact of EV deployment on the load profile of Korean power grid in year 2020 is presented in [89]. The impact of EV charging on the load demand is determined on hourly basis for few scenarios, which include different EV specifications, EV usage patterns, charging rates and charging locations. The paper shows that additional EV loads will increase the load profile and can affect the reliability of the power grid. However, this problem can be solved by the implementation of time-of-use (TOU) tariff system, which shifts the EV loads from peak hours to off-peak periods.

In short, EV deployment will affect the load profile of power grid as EVs are additional loads to be connected to power grid to receive charging. It has a high probability that EV are charged during residential peak load periods because EV owners tend to start charging their EVs once they reach home after work. As a result, large fleets of EV charging will increase the peak load of the power grid load profile [90–95]. Fortunately, many solutions can be implemented to solve this problem such as the implementation of TOU tariff system and appropriate charging management strategies.

4.3.2. Impact on system components

Large fleets of EV charging from power grid requires huge amount of power to be transmitted from power generation plants to these loads. This situation may overload the existing system components because these components may not be designed to cater for the new additional EV loads. Overloading of power grid components, such as transformer and cable can be the major restriction for wider EV

uptake. Numerous studies have been undertaken on this aspect and the main literature is summarized here.

The impacts of EV charging on overhead distribution transformer are examined in [96–98]. The study in [96] investigates the transformer aging due to AC level 1 charging and AC level 2 charging. For the uncoordinated charging scenario, the results show that overhead distribution transformer experiences more aging because of AC level 2 charging than AC level 1 charging. The transformer aging factor could increase up to 8.15 in AC level 2 charging if compared to 3.24 in AC level 1 charging. The study in [97] has similar outcome and concludes that higher penetration rate of EV will exert more negative influence on transformer lifespan. On the other hand, the study in [98] shows that the existing transformers can manage the additional EV demands in most cases. The uncoordinated AC level 1 EV charging has slight effect on the transformer lifespan. However, with high penetration rate of EV under AC level 2 charging level could cause transformer failure due to excessive operating temperature. The impact of EV charging on transformer can be eliminated by off-peak charging and proper load management [96–98].

Some studies investigate the impact of EV charging on the distribution cable or line loading and the recent studies are discussed [99–101]. A real distribution system in Canada is considered in [99] to evaluate the impact of uncoordinated EV charging during peak demand period on cable loading. The results show that the existing cables can only cope with 15% of EV penetration rate for fast charging and 25% of EV penetration rate for normal charging. This study concludes that the present distribution system cannot handle for high EV penetration rates. Study in [100] makes an assumption that the EV fast charging stations will draw power from 20 kV medium voltage (MV) distribution network. The loading of the 400 V secondary cable, which supplies the EV fast charging stations increases tremendously during EV charging. The cable is overloaded and reinforcement is required for EV adoption. The impact of EV charging on the Finnish distribution network is carried out in [101]. The study is performed with the real load profile measurement data, which provided by two Finnish distribution network companies. According to this study, the impact of EV charging on the loading of MV and low voltage (LV) cables are practically imperceptible.

It can be said that the impact of EV charging on system components, particularly the distribution transformer and cable are network specific. Different network configurations, component ratings, component loadings, EV penetration rates and charging strategies will affect the outcome of the power system study. Generally speaking, EVs are additional loads that will increase the loading of system component. Therefore, proper network planning and load management are recommended for future EV adoption.

4.3.3. Impact on system losses

EV charging requires power to be transmitted from power generation plants to EV loads. This transmission of power causes more system losses across the power grid components. This issue raises concerns of power utilities, as they have to bear these additional system losses due to EV charging. Comprehensive literatures have been carried out to investigate the impact of EV charging on system losses.

The impact of EV charging on a typical Danish distribution network is undertaken in [102]. The power system study is performed for the base case without EV integration and an increasing EV penetration rates up to 50%. For the EV adoption rate of 50%, the system losses increase 40% if compared to the base case in the uncontrolled EV charging scenario. Controlled EV charging can reduce the system losses by 10%. Another study in [103] also investigates the impact of EV charging on system energy losses. In the context of system energy losses, EV charging during off-peak hours are considered as the worst case scenario because most of the EVs are

assumed to be charged overnight. The system energy losses increase dramatically in this situation when the EVs are charged from the distribution network. The maximum increase of system energy losses reach up to 40% if compared to the base case without EV integration.

Similar findings are acquired in [92,93, 104] as the adoption of EV into power grid will lead to significant increase in system losses. An optimization objective function to minimize system losses in proposed in [104] to mitigate the negative impact brought by EV charging. Since the accurate forecasting household load is not available, this study introduces stochastic programming to get an optimal coordinated charging load profile with minimized system losses. A study on distribution transformer losses due to residential EV charging is carried out in [105]. A detailed 1200 nodes radial test network is constructed with EV integration at 415 V level. Detailed transformer model is designed and typical Australian residential load profile is used in this study. Different EV penetration rates up to 42% are considered. The result shows that EV charging has obvious impact on distribution transformer losses, which can increase up to 300% for high EV adoption rate.

From the perspective of power grid, EV charging increases the amount of power flow and causes more system losses. In order to reduce the impact of EV charging on system losses, coordinated EV charging can be implemented. Furthermore, supplying EV loads with nearby distributed generation is among the possible approaches to lower the system losses [13].

4.3.4. Impact on voltage profile and phase unbalance

EV charging from power grid will cause voltage drop and voltage deviation on the EV interconnection point. Hence, large fleets of EV charging may make the network voltage violate the safe regulatory voltage requirements. Another impact of EV charging on power grid is phase unbalance, which is contributed by single-phase AC charging.

The Monte Carlo simulation method is adopted in [92] to evaluate the effect of EV charging on the system voltage deviation. Two EV charging strategies are implemented, which are uncoordinated charging and vehicle-to-grid. With reference to the Chinese power grid regulation, voltage deviations up to 7% on the 10 kV power grid are allowable. For uncoordinated EV charging, EV penetration rate of 60% or higher will cause several network voltages to violate the voltage deviation tolerance of 7%. When V2G mode is implemented, all the network voltages are maintained at the acceptable voltage standard with EV penetration rate up to 90%. The reason is V2G mode can achieve load levelling and produces smaller voltage difference between peak and off-peak demands. Similarly, study in [93] shows that EV penetration rate of 50% or higher will cause network voltages to violate the voltage deviation tolerance of 7%. With the implementation of smart charging strategy, all the network voltages are within the acceptable voltage limits.

In contrast, studies in [100, 101] claim that EV charging does not have significant impact on the network voltage. In [100], the outcome of the study shows that all network voltage drop due to EV charging are within the acceptable level, as well as the voltage deviations are not more than 1%. The limiting factor for wide EV adoption is due to system component overloading problem, rather than voltage issue. The study in [101] concludes that EV charging will slightly increase the loading of system components without violating any voltage limits.

The residential EV slow charging may cause severe phase unbalance problem if this kind of charging is not distributed evenly across all three supply phases. Study in [106] connects all EVs to phase “a” to investigate the impact of AC single-phase EV charging on the phase unbalance issue. The result shows that serious phase unbalance problem happens and need extra attentions for future EV deployment. Another study performed in [107] shows that integration of many EV chargers on distribution

network has slight impact on current and voltage unbalance. The phase unbalance remains within acceptable limits over a wide range of tested conditions. A voltage unbalance analysis by using the Genetic Algorithm is performed in [108]. The main purpose of this study is to utilize the proposed Genetic Algorithm to optimize the number of EVs that can be connected at all phases without causing much voltage unbalance.

A large number of researches have been carried out to study the impact of EV charging on voltage drop, voltage deviation and phase unbalance. Some results show that the impacts are significant whereas others show insignificant effects. However, all of these results are influenced by several factors, such as the strength of the test network, EV interconnection point, EV penetration levels, EV charging characteristics and many more. Since EV adoption rate is anticipated to accelerate in near future, voltage regulating equipment and voltage support strategies can be implemented to maintain the voltage profile of distribution network within acceptable limits. Appropriate load management can prevent the phase unbalance problem due to EV slow charging by distributing the EV loads evenly across all three-phases.

4.3.5. Harmonic impact

EV charger uses power electronics for its charging operation. The switching of these power electronics components may bring power quality issues to the power grid. The main concern is the harmonic problem as high harmonic distortion may lead to system components de-rating.

An early paper shows that the increase in voltage THD due to EV charging was less than 1% [109]. In other words, EV charging will not affect the grid power quality in term of harmonic issue. In order to study the harmonic impact of EV charging on power grid, a more comprehensive study is carried out in [110]. This study considers the dynamic behaviors of EV charging, such as random charging time, charging duration and vehicle charging locations. A Monte-Carlo simulation-based method is proposed to perform the system analysis. The study shows similar results where EV charging has negligible harmonic impact on power grid.

On the other hand, authors in [111] show that EVs fast charging injects significant harmonic into power grid. With reference to the EN 50160 standard, the voltage THD of the supply voltage up to 40th harmonics, must not exceed 8%. However, this study reveals that the voltage THD reaches 11.4% with just few EV fast charging. The authors also propose an unique solution to use PV inverter control as active filter to solve the harmonic problem due to EV charging. Similar outcome is shown in other studies, where random EV charging can cause unacceptable level of voltage THD [112,113]. For instance in [113], high penetration of 18 EVs random charging during peak demand hours results in approximately 45% of voltage THD. The uniformly distributed EV charging can noticeably improve the system performance but a more sophisticated coordinated charging scheme is needed.

In short, different studies acquire different outcomes. Some studies show that EV charging has no significant harmonic impact on power grid but others show the completely opposite results. The occurrence of this situation is strongly dependent on how the characteristics of EV chargers are modeled and how the system study is performed. Even though EV charging may introduce harmonic into power grid, many solutions are available to cater for this problem, such as inclusion of filtering device.

4.3.6. Stability impact

Power system stability is defined as the ability of a electric power grid to bring the operation back to steady-state condition after the occurrence of disturbance or transient. A stable power grid is extremely important for reliable power supply. Since EV is the new load to the power grid, the impact of EV charging on distribution network stability is still unknown and should be investigated.

The core finding of many studies states that EV charging significantly reduces the stability of the power grid [114–117]. The study in [114] concludes that power grid with EV integration is more sensitive to the disturbance and perform less stable in term of the time required to reach the steady-state condition. The reason is because the special characteristics of the EV chargers, which inject current harmonics and absorb reactive power. This statement is also supported by [117], which claims that EV load behaviour is special and is represented by a combination of constant power component and negative exponential component. This study investigates the impact of EV charging on power grid voltage stability and concludes that EV charging greatly reduces the steady-state voltage stability of power grid.

The study in [118] evaluates the impact of EV charging on power system stability and propose solutions to solve the problem. EV charging has introduced negative problem to the power system stability. Therefore, a wide-area controller is used to provide auxiliary control signals to the power grid components for power system stability improvement during the EV charging and vehicle-to grid operation. Particle swarm optimization is used to generate the optimized auxiliary control signals to provide the maximum damping to the generators. Most of the literatures investigate the negative impact of EV charging on power system stability. However, a study in [119] has discovered that the integration of EV on the power grid can actually enhance the transient stability of the power grid.

Hence, the integration of EV on the power grid to receive charging can results in certain power system stability issues. On the other hand, EV discharging or vehicle-to-grid operation may enhance the stability of power grid. Therefore, more detailed studies are required for better understanding on this matter. In addition, accurate EV modeling for power system stability study is crucial to get accurate findings.

5. Interconnection of EV in the smart grid

The present transportation sector uses gasoline or petrol for propulsion. There is no interconnection between the transportation sector and power grid during that time. However, the situation changes with the wide adoption of EV into market because EV can be plugged-into power grid to receive energy. Once the interrelation between power grid and transportation sector exists, extensive research have been carried out to study the negative impacts of EV charging on the power grid, which are previously addressed in Section 4.3. Other than challenges, EV deployment can actually bring many opportunities to the parallelly developed smart grid. Some interesting opportunities brought by EV deployment in the smart grid are the vehicle-to-grid technology and integration of renewable energy sources and EV.

5.1. Smart grid

Smart grid is a modernized electrical power grid that utilizes the computer-based remote control and automation to improve the reliability, efficiency and sustainability of the power supply [120]. The inclusion of information and communication technology into the electrical power grid provides two-way communications between utility and customers [121]. Therefore, various components of the system are linked together with bi-directional communication and energy paths to provide greater interoperability between them. Sensors and smart meters are installed throughout the smart grid for real time data acquisition [122]. Along with the collected information, smart grid utilizes autonomous and intelligent monitoring control to supervise and optimize the overall operations of the interconnected components [123].

One of the interesting characteristics of smart grid is that the consumers can actively participate in the grid operation. The

consumers can access the real time information about the electricity usage, tariff and incentive through the advanced metering infrastructure. They can decide their own electricity usage patterns and preferences. By changing their electricity usage patterns, the consumers can help to balance the energy supply and demand [124]. In addition, smart grid uses widely dispersed distributed generation units, which provides better reliability and reduce risk against attacks and natural disaster. Even if problem occurs, a self-healing smart grid will take immediate corrective measures to restore itself from disruption, such as isolates that particular faulty line [125]. Table 5 presents the characteristics of smart grid in comparison to the conventional power grid.

The implementation of smart grid can improve the grid reliability and power quality. There are numerous smart grid projects currently undertaken around the world. With reference to the Global Smart Grid Federation Report [126], some of the leading smart grid projects are the Smart Grid Smart City in Australia, Ontario Smart Metering Initiative in Canada, Low Carbon London in Great Britain, ECAR Project in Ireland, Yokohama Smart City Project in Japan, Jeju Smart Grid System in South Korea and Houston's Smart Grid in United States. Fig. 6 depicts a general framework of smart grid [120,121,124–127]. In comparison to the conventional grid, the crucial feature of a smart grid is the inclusion of the extensive bi-directional communication paths connecting various components. These two-way communication paths have enabled the development of various applications in a smart grid, such as

advanced metering infrastructure, home automation network, demand response, integration of distributed generations and vehicle-to-grid. The smart grid technologies and applications will be further discussed, as follows:

- Advanced metering infrastructure (AMI): Advanced metering infrastructure (AMI) or smart meter is an electronic device that records and collects the real-time information about electricity usage of the customers. The AMI technology enables a two-way communication between power utility and the customers and the communication linkage allows information sharing. The grid operators can utilize the collected information for better monitoring, management and electricity billing purpose [128–130].
- Supervisory control and data acquisition (SCADA): Supervisory control and data acquisition (SCADA) is a centralized system, which utilizes the bi-directional communication paths in a smart grid to collect real-time data from the monitoring devices and control various equipment remotely [121,131]. SCADA is one of the advanced control technologies and has been implemented in many power grids. SCADA consists of a few major components, such as the database servers for data collection and processing from remote terminal units distributed over the system, interface devices connected to sensors, programmable logic controllers, switches and relays and also a two-way communication system for transferring data and control signals [124].
- Home automation network (HAN): Home Area Network (HAN) is an intelligent infrastructure, which enables the communication of different electric appliances within a home. HAN technology can provide automated monitoring, management and control to all connected electric appliances with the purpose of creating a safe, energy efficient and economical living environment. HAN is also designed to operate in collaboration with the AMI, which enables the system to receive the latest real-time information of the market and power grid conditions. This function helps the HAN system to respond and control the connected electric appliances according to the customer preference and real-time grid condition [132–134].
- Demand response: Demand response is a smart grid technology that encourages the customers participation in improving the operation of power grid, and in return, the customers will be rewarded with incentives. Customers can reduce their electricity

Table 5
Comparison of conventional grid and smart grid [120–125].

Characteristics	Conventional grid	Smart grid
Communication	Uni-directional	Bi-directional
Monitoring control	Manual	Autonomous and intelligent
Inclusion of smart sensors and meters	Limited	Throughout
Consumer participation	Passive	Active
Power generation	Centralized	Distributed
Recovery	Manual	Self-healing

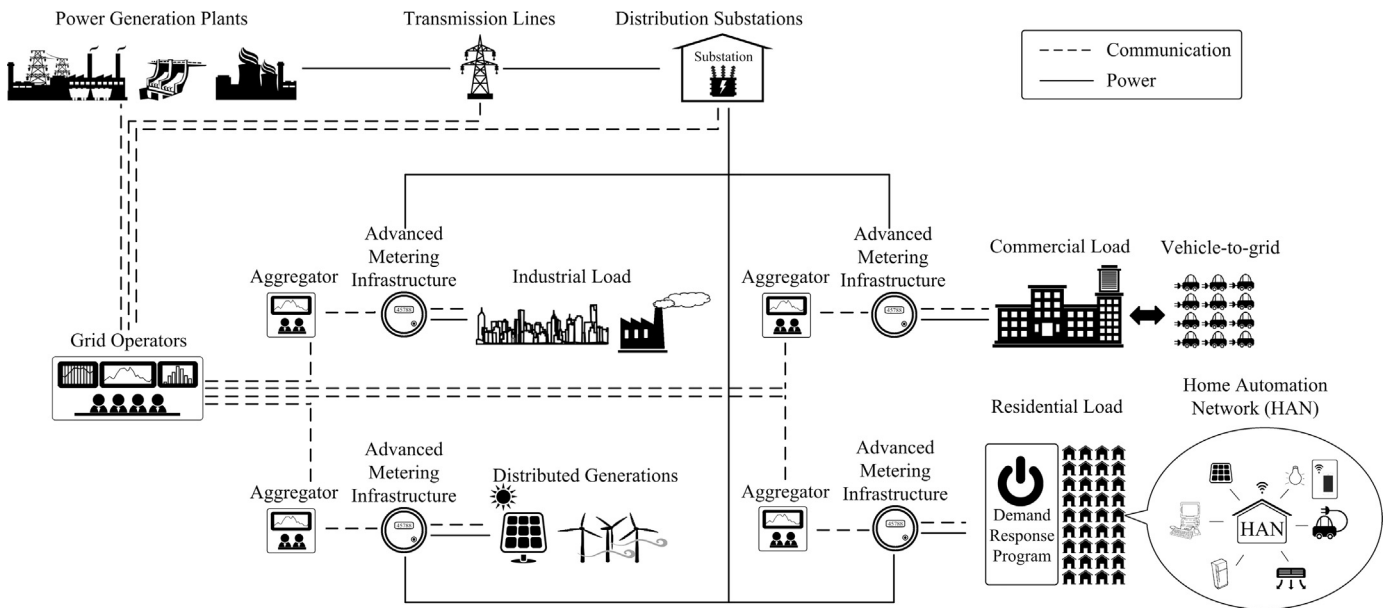


Fig. 6. Smart grid framework [120,121,124–127].

usage during emergency or peak load period in response to reduce the power grid load. Hence, demand response can effectively avoid additional power generation to meet the peak load demand or emergency situation, which will be a cost-effective alternative to sustain the power grid operation [135–137].

- Integration of distributed generations: In a smart grid, power generation units are widely distributed across the whole power grid, instead of using several major power plants at generation level [131]. The integration of distributed generations allows the consumers at distribution level to generate power and supplies to nearby loads. This can reduce power losses for the smart grid since power does not required to be transmitted from the upstream generation plants. The concept of distributed generation has encouraged the wide adoption of renewable energy sources, especially the solar photovoltaic and wind turbines. The renewable distributed generation can provide a green and sustainable electrical power grid. In addition, storage technologies can also be integrated in the smart grid as supplementary to the renewable distributed generation to store the excessive generated power for later use [124].
- Vehicle-to-grid (V2G): With the bi-directional communication infrastructure, the implementation of vehicle-to-grid (V2G) concept will become feasible by controlling and managing the energy exchange between the power grid and EV battery [121]. EV receives charging from the power grid whenever the charge level of EV battery is low. In V2G operation, the charge level of the EV battery is continuously monitored and allowed to be discharged to the smart grid. EVs can be considered as dynamic distributed energy storages. Therefore, proper V2G management of a large fleet of EV are important to achieve various benefits, such as active power regulation, reactive power regulation and ancillary service support [138].

5.2. Prospects of EV deployment in smart grid

Due to the characteristics and benefits of smart grid, power utilities around the world have put efforts in improving the conventional power grid to smart grid. The advancement of smart grid technologies have reached to certain maturity level, which creates more opportunities and fosters new applications. Along with the recent EV deployment, the improved smart grid technologies have promoted the vehicle-to-grid (V2G) technology. Furthermore, the implementation of V2G technology in the smart grid system facilitates the deployment of renewable distributed generation for mutual benefits.

5.2.1. Vehicle-to-grid (V2G) technology in smart grid

The development of V2G technology has gone through a few phases. Uni-directional V2G is the early stage of the V2G technology implementation because uni-directional V2G can be achieved using the existing standard EV chargers with additional communication feature [139]. In addition, the realization of unidirectional V2G technology is inexpensive and does not degrade the EV battery from discharging since it only controls energy flows from the power grid to EV [140]. Uni-directional V2G technology manages the EV charging process by controlling the charging rate based on certain energy scheduling or incentive system. The introduction of an energy scheduling policy is the main driver to accomplish the uni-directional V2G technology [141,142]. The EV owners will be rewarded with incentives when they charge their EVs during off-peak period and limit EV charging during peak hour. The controlled EV charging will disperse the EV loads over the off-peak periods and level the load profile for cost saving and power losses reduction [139,140]. Therefore, the implementation of the uni-directional V2G technology can attain mutual benefits for EV owners and power grid.

Recently, bi-directional V2G technology received more attention due to the development of the bi-directional EV chargers, which have enabled the two-ways communication and energy exchange between EVs and power grid. The bi-directional V2G technology can accomplish many benefits based on the control schemes and preset objectives [127]. Due to the importance of bi-directional V2G technology, from here onwards, the term V2G is meant to represent the bi-directional V2G technology. V2G technology can be classified into several categories based on the scale of EV fleet, such as vehicle-to-home (V2H), vehicle-to-vehicle (V2V) and vehicle-to-grid (V2G) [143–145]. V2H has the smallest scope among all the categories since its application is within a home automation network. In a home automation network, EV can act as an energy storage to store the excessive power generation from the home solar photovoltaic unit and discharge to supply the other smart appliances whenever the home renewable generation is low. In collaboration with the renewable distributed generation, V2H technology is able to create a safe, energy efficient and environmental-friendly living environment [143].

In a larger EV fleet context like EV parking lots at commercial building, the role of the EV aggregator become significant to control and manage the energy exchange of all EVs to meet certain objectives. V2V technology allows energy sharing among EVs [144]. Meanwhile, the V2G technology can transfer energy from the power grid to charge EVs, as well as discharge EVs batteries charges into power grid for grid support [145,146]. With appropriate optimization control and management system, V2G technology can bring various benefits to the interconnected power grid, such as peak shaving, load leveling and voltage regulation. Fig. 7 depicts a framework of V2H, V2V and V2G technologies with the integration of renewable distributed generations.

V2G technology can bring various benefits to the power grid via proper V2G control and management. However, the large fleet of

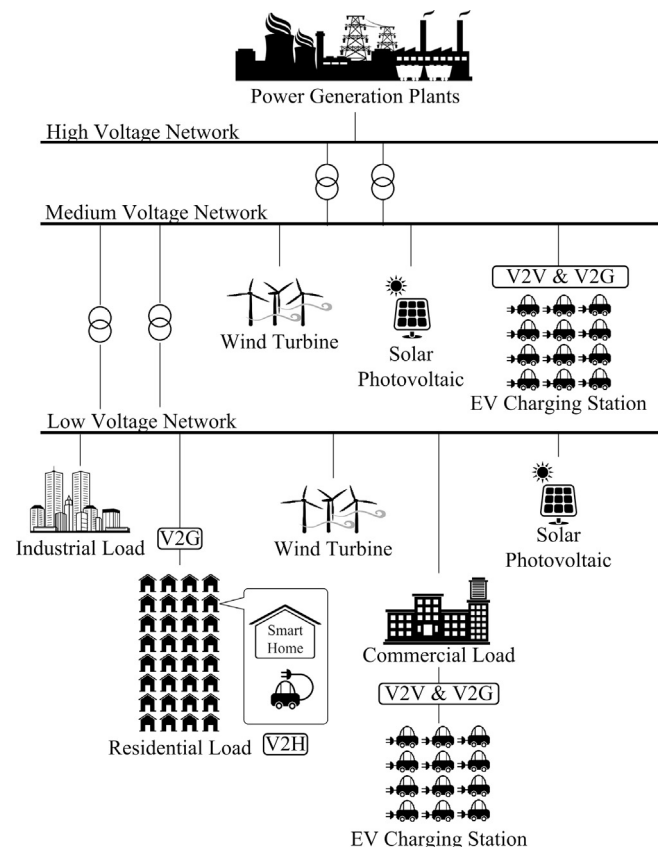


Fig. 7. Framework of V2H, V2V and V2G technologies with renewable-based distributed generation [143–146].

grid-connected EVs will result in many uncertain constraints to the power grid, such as different state of charge levels of EV batteries and the dynamic probability of EV connection. In order to manage the large fleet of EVs, unit commitment optimization technique is used for planning and controlling the energy flow between EVs and power grid. With the optimization of V2G algorithm, various power grid benefits can be accomplished based on the predefined objective functions. Some of the benefits accomplished by the V2G technology are listed, as follows:

- **Ancillary service – spinning reserve:** Ancillary service refers to the supporting service supplied to the power grid for the purpose of maintaining the reliability and sustainability of the power grid [147]. V2G technology provides ancillary service to the power grid via spinning reserve service, where the energy stored in the grid-connected EVs is used as the additional generation capacity to compensate the generation outage [148]. With the spinning reserve service provided by V2G technology, it can provide failure recovery, as well as reduce the backup generation capacity [149, 150].
- **Active power support/load leveling and peak load shaving:** It is a common phenomenon that a typical commercial or residential load profile has the peak load for only a short period. It is desirable to level the load profile to prevent the aging of power grid equipment and grid overloading, as well as to achieve energy efficiency and economic benefits [151]. V2G technology is able to achieve these benefits by utilizing the excessive EVs energy to provide active power support to the power grid during peak hour and charge the EVs during off-peak hour. These techniques are denominated as the “peak load shaving” and “load leveling”, respectively [152,153].
- **Reactive power support/power factor correction/voltage regulation:** Large scale of EV charging is a massive challenge to the power grid. Power efficiency and voltage stability are important power qualities to be regulated throughout the power grid operation for grid reliability [154]. The conventional method of voltage regulation and power factor correction is by the installment of the static volt-ampere reactive compensator to supply reactive power support to the power grid [155]. With the implementation of V2G technology, the reactive power compensation service for grid voltage regulation and power factor correction can be accomplished with the grid-connected bi-directional EV chargers. The DC-link capacitor

connected in a bidirectional EV charger is able to provide reactive power support to the power grid with an appropriate switching control for the EV converter [156,157].

- **Harmonic filtering:** The modern power grid consists of many high power non-linear loads, which generates significant amount of current harmonics into the power grid. EV chargers are one of the harmonic source due to the use of converter switching [158]. Therefore, the implementation of V2G technology will affect the power quality of the smart grid if no corrective measure is taken. Other than the inclusion of additional filter device to solve the harmonic problem, EV chargers with appropriate control can be used as the active filter to filter out the harmonics generated by EV chargers and other non-linear loads [158–160]. The converter of the EV charger can operate as variable impedance for each individual harmonic frequency and solve the harmonic problem with proper filtering strategy [160].
- **Support for the deployment of renewable energy resources:** With the implementation of proper V2G control strategy, EVs can be used as the solution for the renewable energy intermittency issue [161–164]. EVs will charge from the power grid when renewable distributed generation generates excessive power and discharge to power grid when renewable distributed generation does not generate enough power. In addition, EV battery can be used as the energy storage system to regulate the voltage of a power grid with renewable energy integration [163]. Since V2G technology can be utilized to solve the intermittency issue of renewable energy sources, more renewable energy sources can be integrated into power grid to achieve a more sustainable electrical power system [164].

Table 6 shows the comparison of the uni-directional V2G and bi-directional V2G technology. The services, benefits and drawbacks of each V2G technology are summarized. The implementation of V2G technology in a smart grid can bring numerous possible services, which are advantageous to the grid operation. It is important to mention that the implementation of V2G technology in the smart grid system facilitates the integration of renewable distributed generation while EV charging from renewable distributed generation will be more environmentally-friendly. The interaction of EVs with renewable energy resources can achieve mutual benefits and therefore, will be further discussed in the next section.

Table 6
Comparison of uni-directional V2G and bi-directional V2G.

V2G Types	Description	Services	Benefits	Drawbacks	References
Uni-directional	Uni-directional V2G technology controls the EV charging rate in a single power flow direction from the grid to EV based on energy scheduling and incentive system.	(a) Ancillary service - load levelling	(a) Minimize power losses (b) Maximize profit (c) Minimize operation cost (d) Minimize emission	(a) Limited service available	[139–142]
Bi-directional	Bi-directional V2G refers to the dual direction power flow between EV and the power grid to achieve numerous benefits.	(a) Ancillary service - spinning reserve (b) Active power support / load leveling and peak load shaving (c) Reactive power support / power factor correction/ voltage regulation (d) Harmonic filtering (e) Support for the integration of renewable energy resources	(a) Minimize power losses (b) Maximize profit (c) Minimize operation cost (d) Minimize emission (e) Prevent power grid overloading (f) Improve load profile (g) Regulate voltage level (h) Failure recovery (i) Maximize renewable energy generation	(a) Battery degradation (b) Complex hardware infrastructure (c) High investment cost (d) Social barriers	[147–164]

5.2.2. Interaction of renewable energy sources (RES) with V2G in smart grid

Renewable energy sources (RES) are widely adopted across the world for their clean and free supplies. The adoption of renewable energy into power grid will create a greener and more sustainable electrical power system. The primary RES are wind energy and solar photovoltaic. Despite the attractive environmental advantage, the power generation of RES is intermittent and heavily depends on the weather situation. The variation of wind speed and solar irradiance have resulted in fluctuation in power generation for wind turbine and solar photovoltaic, respectively. The intermittency issue of RES becomes the main barrier for the RES deployment. The utilization of the stationary energy storage system to absorb excessive RES generation or supply energy in the case of low RES generation has been proposed in the literature to cater for the problem [127]. However, this solution for the RES intermittency issue requires high investment cost, which causes the delay of RES deployment.

The recent EV roll out has revolutionized the power grid. EVs can be considered as distributed energy storage because EVs can charge power from and discharge power into power grid via V2G technology. Hence, EVs energy storages can be used to stabilize the inconsistent generation of RES and accommodate more RES integration. On the other hand, EV charging from the power grid with high penetration of RES will enjoy low wells-to-wheels emissions. Therefore, the interaction of EVs with RES can achieve mutual benefits enabling the power grid towards sustainability. Extensive literature has been emphasized on the integration of V2G technology and the RES in smart grid system. Table 7 summarized the recent literature on the interaction of RES with EV for smart home, parking lot, distribution system and micro-grid.

5.2.2.1. Interaction of EVs with solar photovoltaic. Various research have been carried out to access the feasibility of EV interaction with solar photovoltaic in the power grid, ranging from small-scale systems like smart home and parking lot to larger systems, such as distribution network and micro-grid. In the context of smart home,

the impact assessment on the emissions reduction for a home energy management system comprised of solar photovoltaic and EVs has been carried out in Japan [165]. The research shows that the greenhouse gas emissions are greatly reduced with both photovoltaic generation and EV technology, which shows that the interaction of EV with solar photovoltaic have positive influence in emissions reduction. The development, modeling and sizing of a standalone solar-based home EV charging station is proposed in [166]. By using the proposed solar-based charging station, the authors propose various EV charging control topologies for overnight charging but only limited to home-to-vehicle application. In [167], a comprehensive study about the development of an uninterruptible future home system by utilizing the interaction of V2G technology with photovoltaic generation is presented. The proposed home system can island from the faulted power grid, sustain in the standalone mode and reconnect back to the power grid without any interruption. The uninterruptible future home concept is realized by the grid-interactive EV and solar photovoltaic, which can provide voltage and frequency regulation in the islanded standalone mode.

Other than smart home, there are some studies being performed to investigate the viability of EV interaction with solar photovoltaic in the context of parking lot. The impact analysis of EV charging using solar energy at workplace parking lot are investigated in [168,169]. In [168], the authors perform the energy economic and carbon emissions analysis and concludes that EV charging is best suited to daytime charging with solar energy at workplace parking lot. The authors in [169] propose an EV charge scheduling strategy to investigate the payback period and carbon emissions of the solar-based EV charging station at workplace parking lot. The results show that the EV charge scheduling strategy has slight influence on the payback period of the proposed parking system but can significantly reduce the emissions by 90% compared to normal EV charging station. From the economical and technical perspectives, the authors in [170,171] investigate the interaction of stochastic EV charging and discharging scheduling with solar photovoltaic in an intelligent parking

Table 7
Interaction of RES with EV in various power system.

Interaction of RES with EV	Application scopes	Contributions	References
Solar photovoltaic and EV	(a) Smart home	(a) Implication of EV and photovoltaic deployment in smart home system for emission reduction. (b) Development of standalone solar-based home EV charger for home-to-vehicle application. (c) Development of uninterruptible future home by utilizing the interaction of solar photovoltaic with V2G technology.	[165] [166] [167]
	(b) Parking lot	(a) Impact analysis of EV charging using solar energy at workplace parking lot. (b) Stochastic charging and discharging scheduling for EV intelligent parking lot consisting of solar photovoltaic.	[168,169] [170,171]
	(c) Grid distribution network	(a) Impact assessment on the power system performance with the integration of grid-connected EVs and photovoltaic. (b) Design of EV charging control strategy for a grid-connected solar-based charging station. (c) Development of optimization algorithm for coordination of V2G service and solar photovoltaic.	[172–174] [175,176] [163,177]
	(d) Micro-grid	(a) Generation scheduling formulation for industrial micro-grid consisted of solar photovoltaic generation and EVs.	[178]
Wind turbine and EV	(a) Grid distribution network	(a) Research on the potentials of EV interaction with wind energy generation in the power system. (b) Development of V2G optimization scheme to solve wind intermittency issue.	[179,180] [181,182]
	(b) Micro-grid	(a) Coordinated energy dispatching of V2G technology and wind generation via optimization algorithm.	[183]
Solar photovoltaic, wind turbine and EV	(a) Smart home	(a) Design of control strategy for a smart home consisting of renewable energy sources and grid-interactive EV.	[143]
	(b) Parking lot	(a) Development of intelligent optimization framework for the integration of EVs and renewable energy resources.	[162]
	(c) Grid distribution network	(a) Potential assessment of grid-connected EVs in balancing the intermittent generation of renewable energy sources. (b) Analysis of EV emissions associated with the renewable energy source generation. (c) Optimized algorithm to integrate grid-connected EVs and renewable energy resources.	[184] [185] [161,186]
	(d) Micro-grid	(a) Propose of V2G control to maximize renewable energy sources integration in micro-grid.	[164,187]

lot. Many practical constraints have been taken into consideration in the optimization algorithm, such as battery lifetime, battery state-of-charge, remaining charging time, irradiance probability, spinning reserve requirement and preferred charging price. The optimization algorithm provides appropriate V2G energy management to maximize the EV charging and discharging rates, maximize the state-of-charge of EVs batteries, minimize the increased demand during peak load period, maximize the incentives to participated EV owners and stabilize the intermittent generation of solar photovoltaic. The results show that the proposed V2G energy management system at solar-based parking lot satisfies both financial and technical objectives [170,171].

Within the framework of the distribution network, many literature have investigated the practicality of co-existence between solar photovoltaic and EVs. One recent study examines the effect on the power grid performance with the integration of grid-connected EVs and photovoltaic in the residential distribution system [172]. The study reveals that solar generation can only meet the EV charging demand for short-term without any system upgrade. However, in the long-term, the solar generation will bring obvious reverse power flow issue and will not be able to meet the demand of high EV penetration. In [173], the authors conclude that EVs and solar photovoltaic are feasible to work together. Solar photovoltaic generates high power during daytime and can be used to supply the daytime charging demand of EVs. This concept allows EV charging throughout the whole day without causing increase in peak load demand or overloading problem. Meanwhile, EVs can absorb the excessive photovoltaic energy during low demand period and used to solve the intermittency issue of solar photovoltaic. The authors in [174] investigate the impact of EV and solar photovoltaic deployment on the performance of distribution system. This study shows that photovoltaic deployment can be used to supply one-fifth of the total electricity load, which indirectly reduce the power generation of other conventional sources. In addition, EV deployment can reduce the peak load demand by utilizing the V2G technology. Therefore, both EV and solar photovoltaic deployment can prevent the overloading problem in the distribution network.

In the context of the distribution system, some research are conducted on the design of EV charging control strategy for the solar-based charging station. A smart EV charging station integrated with solar photovoltaic and power grid is proposed in [175]. The proposed system consists of the solar photovoltaic with DC/DC boost converter, EVs with DC/DC buck converter and the grid with DC/AC bi-directional converter. An unique DC-link voltage sensing technique is proposed to control the energy flow. The development of the control strategy has the main objective to charge EV using minimum energy from the grid for better grid stability, energy efficiency and grid asset utilization. Laboratory prototype has been developed to validate the effectiveness of the proposed EV charging control strategy for the solar-based charging station. Likewise, a similar study is conducted in [176] to propose an optimal EV charging control strategy for the solar-based charging station. The slight difference of this study compared to the one in [175] is that the EV converter is designed as bi-directional EV charger. Therefore, EV not only can receive charging from the solar photovoltaic and grid, but also able to stabilize the fluctuation in photovoltaic power generation.

The development of optimization algorithm for the coordination of V2G service and solar photovoltaic in the distribution system is performed in [163,177]. The authors in [180] propose a controlled V2G energy management system, which is able to optimally reduce the intermittency issue of solar photovoltaic and increase the revenues for both photovoltaic suppliers and EV owners. On the other hand, a novel method based on mixed integer linear programming is proposed in [163] to utilize the EV public charging station with energy storage in order to regulate the voltage of a low

voltage feeder with high penetration of solar photovoltaic. The appropriate sizing of the energy storage is further investigated to optimize the overall system. The results show that the proposed system architecture and control strategy can effectively provide voltage regulation to the low voltage feeder with photovoltaic sources. In the framework of micro-grid, a generation scheduling formulation for industrial micro-grid, which consists of solar photovoltaic generation and EVs is proposed in [178]. An optimization technique is developed based on dynamic optimal power flow with various constraints from factories, solar photovoltaic storage and EVs dynamic charging. The main objective of this research is to minimize the overall cost of the proposed industrial micro-grid.

In summary, extensive literature have demonstrated the feasibility of the integration of EVs and solar photovoltaic in various power grid contexts. For the interaction of EV with solar photovoltaic, the major research scopes are the impact assessment on the system performance, the development of EV charger controller and energy management optimization strategy. All the literature have concluded that the interaction of EV with solar photovoltaic are beneficial. From the environmental standpoint, both solar photovoltaic and EVs are green technologies that can significantly reduce carbon emissions and fossil fuel usage. From the economic perspective, the appropriate energy management of EV and photovoltaic can increase the revenues for both photovoltaic suppliers and EV owners. From the technical viewpoint, the interaction of EV with solar photovoltaic can achieve mutual benefits. EVs can be utilized to absorb the excessive solar generation and supply power to other loads during low solar generation. Meanwhile, EV owners are rewarded with incentives by participating in the V2G technology.

5.2.2.2. Interaction of EVs with wind turbine. Wind energy generations have been broadly developed due to their green nature and economical advantages. Along with the deployment of EV industry, many research associated with the integration of EVs and wind energy have been carried out. From the context of power distribution network, the assessment of future wind energy utilization for EV charging is conducted in Germany and northeastern Brazil [179,180]. In [179], the research shows that there are about 15% of excessive wind power generation by the year 2030 in Germany, which can be utilized for EV charging purpose. The excessive wind power is able to support up to 50% of the EVs load demand. The study in [180] shows the potential of the combined strategy by developing EV market and wind energy generation together in the northeast of Brazil. In the study, it can be concluded that the short-term interaction between EVs and wind energy generation is able to boost the penetration of EVs into market. Meanwhile, EVs can assist the regulation of wind generation intermittency in the long run.

The energy management between the large fleet of EVs and wind turbine-based distributed generators in power distribution system involves various uncertainties and constraints. Therefore, optimization technique is proposed in [181,182] to achieve the maximized benefits in the predefined objectives. In [181], a decentralized EV charging optimal control scheme is proposed. The proposed control strategy has the capability to mitigate the intermittency issue of wind energy generation and maximize the financial benefits for EV owner and power utility simultaneously. An optimized smart EV charging control is developed in [182], which is able to provide ancillary service to the power grid, to reduce the power grid operating cost and compensate the wind generation fluctuation. The outcome of this research can be useful to perform the long-term cost benefits analysis and policy making for the EV interaction with wind energy deployment.

A smart micro-grid system needs to operate independently and reliably with its own power generation and regulation. In a micro-grid, EVs can improve the flexibility of energy management due to

their bi-directional capability. The study in [183] shows the energy dispatch coordination of V2G technology and wind generation via optimization algorithm. There are several energy dispatch coordination methods proposed in this research, which include the valley searching, interruptible and variable rate dispatching methods. The research emphasizes the potential of EVs to absorb and supply energy in order to optimize the energy dispatching within the micro-grid for solving the wind energy intermittency whilst meeting EV users requirements.

In short, several studies have been conducted to investigate the EV interaction with the wind energy generation, especially in the context of power distribution system. The appropriate energy management system will facilitate the integration of EVs and wind generation in the power grid to achieve mutual advantages. The controlled V2G strategy can help to solve the fluctuation of wind generation, especially to absorb the excessive wind generation during low demand periods. In the meantime, EV owners will be compensated with incentives for their participation in the V2G services.

5.2.2.3. Interaction of EVs with both solar photovoltaic and wind turbine. The previous sections demonstrate the interaction of EVs with either solar photovoltaic or wind turbine in the power grid. In fact, a number of studies have been conducted to examine the potential integration of EVs, solar photovoltaic and wind turbine altogether in the power grid. The design of the control strategy for a smart home, which consists of solar photovoltaic, wind generation and grid-interactive EV is presented in [143]. The control strategy is developed to minimize the total energy cost for the proposed smart home system. The results shows that the interaction of EVs with both the RES can achieve 20% saving in the total energy cost. In the context of the parking lot, the authors in [162] have developed an intelligent optimization framework for the integration of EVs and RES. The proposed intelligent optimization framework considers the optimal placement of the system, the optimal EV charging rates and the optimal sizing of the RES. The optimization is performed using Genetic Algorithms and the outcome of this research will be the design steps of a practical grid-friendly EV parking lots.

In the distribution system, a case study to evaluate the potential of grid-connected EVs in balancing the intermittent generation of RES is conducted in [184]. The study shows that smart EV charging significantly improves the generation fluctuation of RES in the distribution system. The investigation on EV emissions associated with the wind and photovoltaic electricity is performed in [185]. The results show noteworthy reduction in greenhouse gas emissions when EVs are charged from wind and photovoltaic electricity. The development of an optimization algorithm to integrate grid-connected EVs and RES in the distribution system is presented by the authors in [161,186]. In both studies, the Lyapunov optimization is utilized to schedule the EV charging with the aims to solve the intermittency issue of RES, improve energy efficiency and reduce the energy cost. Likewise, the authors in [187] formulate a stochastic optimization algorithm for the energy scheduling of EVs, photovoltaic and wind in a micro-grid system. The proposed energy scheduling minimizes the system operational cost, minimizes the power losses and addresses the intermittency of wind and solar generations. In [164], the V2G controller is designed in MATLAB software and is utilized to maximize the RES integration in micro-grid.

In brief, the interaction of EVs with solar photovoltaic and wind turbine are feasibly implemented in the power grid for various benefits, such as energy cost saving and emissions reduction. Moreover, EVs can be utilized to maximize the RES integration in power grid by regulating the intermittent behaviour of solar photovoltaic and wind turbine. The proper V2G implementation allows EVs to absorb the excessive generation of both solar photovoltaic and wind

turbine, while supplying the energy from the batteries to other loads during low renewable energy generation.

6. Research limitations section

EV deployment can bring various benefits to the environment and power grid. From the environmental standpoint, EVs are cleaner than the conventional internal combustion engine vehicles because EVs have zero tailpipe emissions. From the perspective of power grid, EV deployment can bring many opportunities to the smart grid, such as V2G technology and as facilitator to RES integration. Extensive literature has demonstrated that the V2G interaction with RES in the smart grid can bring economical and environmental advantages, as well as provide numerous services and regulations to the power grid. Nevertheless, there are still many barriers, challenges and limitations need to be overcome before the successful EV deployment.

At the present time, EV technologies are not fully-matured. Despite the notable improvements in the past decades, the current lithium-based EV battery has restricted energy density, limited life cycle and high initial cost. There are some battery technologies holding high potential to give superior performance, but still in the experimental phase. Furthermore, utilizing EV battery in the V2G technology will increase the EV charging and discharging cycles, which will accelerate the battery degradation. Therefore, more research needs to be conducted to improve the technical and economical performances of EV battery.

Apart from the battery technology issue, the current EV charger technology has limitations and not ready for the V2G implementation in smart grid. Most of the EV chargers adopted in the market are the uni-directional charger type, regardless for slow charging or fast charging. However, a bi-directional EV charger is required for the implementation of V2G technology and this type of charger is still under experimental stage. Hence, the design and development of the bi-directional chargers needs to get more attentions. In addition, a complete EV charging network with sufficient number of installed EV chargers is necessary for EV deployment but the installation of such complex infrastructure needs proper planning and huge investment cost. Comprehensive investigation need to be conducted on this research scope.

The V2G technology is a significant prospect brought by the EV integration in the power grid, which can bring many services to the power grid and also maximize the RES integration. The realization of V2G technology needs the active participation of EV owners. However, EV owners will be concerned about the disadvantages in participating in the V2G program, such as the battery degradation issue. If EV owners are unwilling to participate, this will be a huge social barrier for the implementation of V2G technology. Therefore, research need to be performed to discover the suitable solution, such as introducing the V2G energy management strategy with appropriate incentive-based policy.

7. Conclusions

This paper comprehensively reviews the current status, impacts and opportunities of EV deployment, as well as the latest development of EV technologies. The global EV outlook looks very promising as the stock of EV was more than 180,000 at the end of 2012. The implementation of incentive-based policy to EV purchase cost, development of charging infrastructure and increased public awareness on environmental issue are among the facilitators for wider EV uptake. Attentions have been placed on the development of EV technologies. Various power train configurations, new battery technologies and different charger converter topologies are

introduced to achieve specific objectives. The paper has also summarized the impacts of EV roll out on the environment, economic and power grid. Without proper EV charging management, EV deployment could exert negative impacts on the power grid. Other than challenges, EV deployment can actually bring many potential opportunities to the smart grid, such as vehicle-to-grid technology and as the support for the renewable energy resources integration.

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